Protein Analysis

function [4]

interactive dashboards:
- → show results so that everyone can understand and handle the data

protein structure
- → understand how mutations, modifications and ligand binding affect protein function

Visualisations

protein sequence alignment [1]
- → channels: colour and position
- → common feature: list all sequences, emphasize differences
- → main problem: big amount of data

phylogenetic trees [2]
- → channels: position, sometimes colours
- → common features: leaves = taxa, nodes show evolutionary relationship
- → main problem: big amount of data
- → aim: show evolutionary relationships

protein topology [3]
- → channels: colour
- → common features: display amino acid sequence, different colours ⇒ different parts of a protein
- → interactive: information about each amino acid
- → aim: show topology (closely related to function)

how

to preserve
- → interactive tools to show more information
- → colour usage
- → different tools for different information

to avoid
- → visualisation of all sequences
  - ⇒ difficult to understand
  - ⇒ solution: preprocessing
- → too many colours
  - ⇒ not clear what they stand for
  - ⇒ solution: less colours
- → only presenting result of analysis
  - ⇒ combine different tools